AI Adoption in Nigerian Firms:
Evidence from Survey and Labor Market Data

Ebehi Iyoha[a] Abdoulaye Ndiaye[b] Chukwuka Onyekwena[c]

[a]Harvard Business School

[b]New York University (Stern)

[c]Center for the Study of Economies of Africa

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Why Study AI Adoption in Africa?

Research Gap:

- Limited systematic examination of AI adoption in Sub-Saharan Africa
- African firms excluded from global AI surveys (e.g., OECD, Stanford HAI)
- Most evidence from developed economies

Policy Urgency:

- Risk of new digital divides
- Opportunity for leapfrogging
- Need for evidence-based AI strategies

African Context

- Productivity gaps vs. innovation potential
- Infrastructure constraints vs. mobile-first solutions
- Young workforce vs. skills mismatches

Research Questions

1. What is the **scope and scale** of AI adoption by Nigerian firms?
2. What **complementary investments** are associated with successful adoption?
3. How does **ROI uncertainty** shape adoption decisions—and is this a firm-level or sector-wide phenomenon?
4. What are the implications for **productivity convergence** vs. divergence?

Data

1. Firm Survey Data

- **N = 552** businesses
- **3 cities:** Lagos (231), Abuja (179), Port Harcourt (142)
- **Sectors:** Services (42%), Manufacturing (23%), Trade (19%)
- **Size:** Micro to Large enterprises

2. Revelio Labs Data

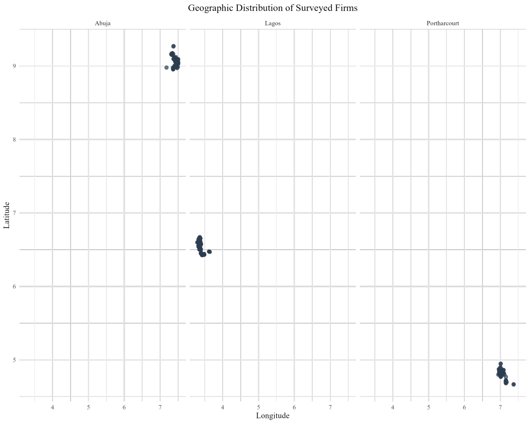
- **7.2M+** African CVs (2021-2024)
- **22,761** AI-skilled workers in Nigeria
- **Matched** to firm characteristics
- **Wage** and skill premium analysis

Complementary approach: Survey captures adoption breadth; Revelio reveals depth and quality

Survey Design and Implementation

Survey Modules:

- AI/digital tool awareness and usage
- Implementation barriers
- Training and infrastructure
- Data governance practices
- Ethical considerations
- Performance impacts



AI Skill Identification in Revelio Data

AI Keywords (LLM-enhanced list)

Machine Learning, Deep Learning, TensorFlow, PyTorch, Keras, Scikit-Learn, NLP, Computer Vision, OpenCV, HuggingFace, Transformers, BERT, GPT, LLMs, MLOps, Prompt Engineering, Data Science, AWS SageMaker, Azure AI, Google Cloud AI, Neural Networks, CNN, RNN, LSTM, GAN, XGBoost

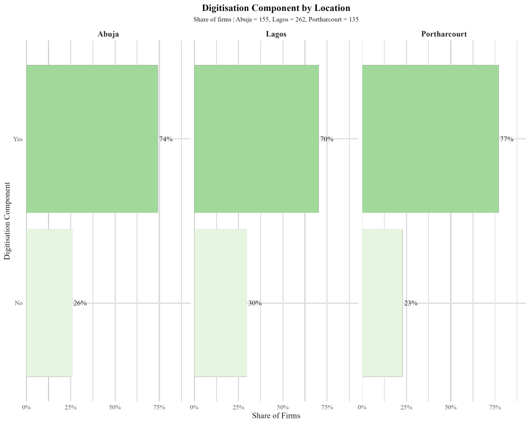
Matching Process:

- Skills extracted from CV text
- Matched to company records
- Industry classification (RICS)
- Domestic vs. foreign firm identification

Key Metrics:

- AI workforce share
- Salary premiums
- Skill concentration
- Seniority distribution

High Adoption, Low Intensity Paradox



Key Findings:

- **74-77%** of firms use at least one AI/digital tool
- But **76%** rate usage intensity as "low/moderate"
- Only **12%** describe usage as "high" or "very high"

Interpretation

Evidence of experimental adoption rather than systematic integration

AI Tool Landscape

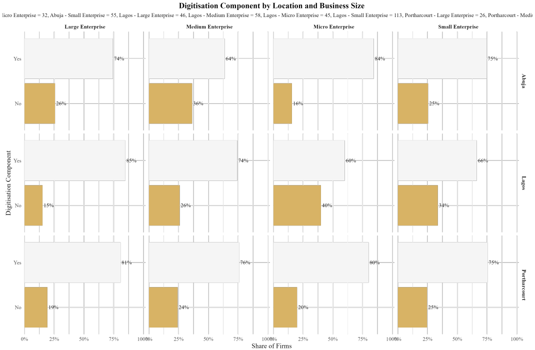
Lagos: Innovation Hub

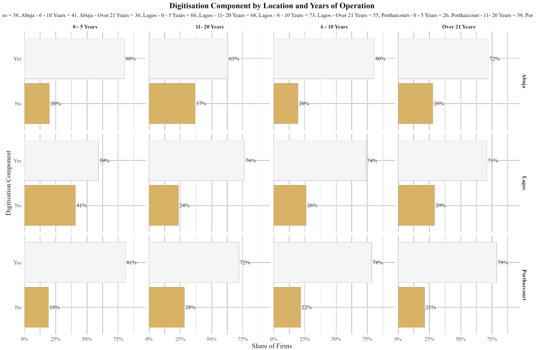
Abuja: Government & Services

- **ChatGPT** dominates (40-55% of adopters)
- Business tools: QuickBooks, chatbots, Canva
- Lagos shows more diverse, cutting-edge tools

Functional Areas of AI Application

Adoption by Firm Size and Age





Key Insights:

- Large firms lead in adoption (85% vs. 68% for micro)
- Young firms (0-5 years) show highest adoption rates
- Suggests both resource advantages and legacy constraints

The Complementarity Hypothesis

AI adoption $\neq$ AI value creation

Human Capital

- Skills training
- AI literacy
- Change management

Infrastructure

- Computing resources
- Data systems
- Network capacity

Organization

- Governance structures
- Process redesign
- Strategic alignment

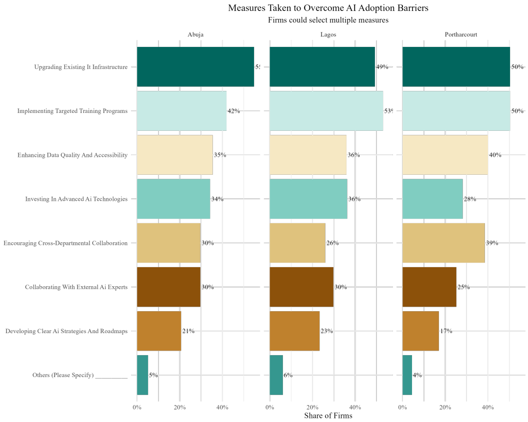
Evidence of Complementary Investments

Among AI-adopting firms:

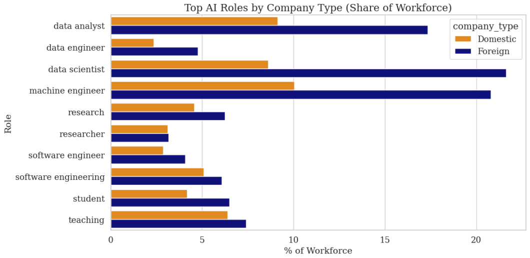
- **58%** provide employee AI training
- **45%** implement data governance
- **36%** have formal AI strategies
- **31%** upgraded IT infrastructure

Size matters:

- Large firms **2.3x** more likely to make complementary investments
- Resource constraints bind for SMEs



AI Workforce Characteristics



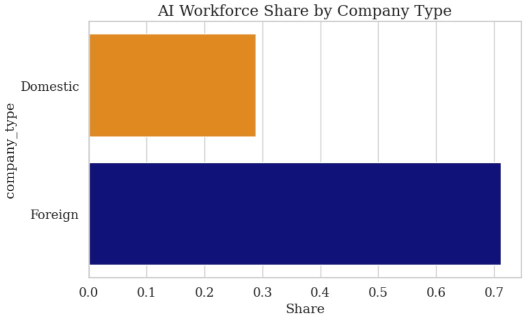
Key Findings:

- **67%** of AI workers are mid/senior level
- Only **11%** are junior employees
- Top roles: Data Scientists, ML Engineers, Business Analysts

Implication

AI adoption requires experienced talent, not just technical skills

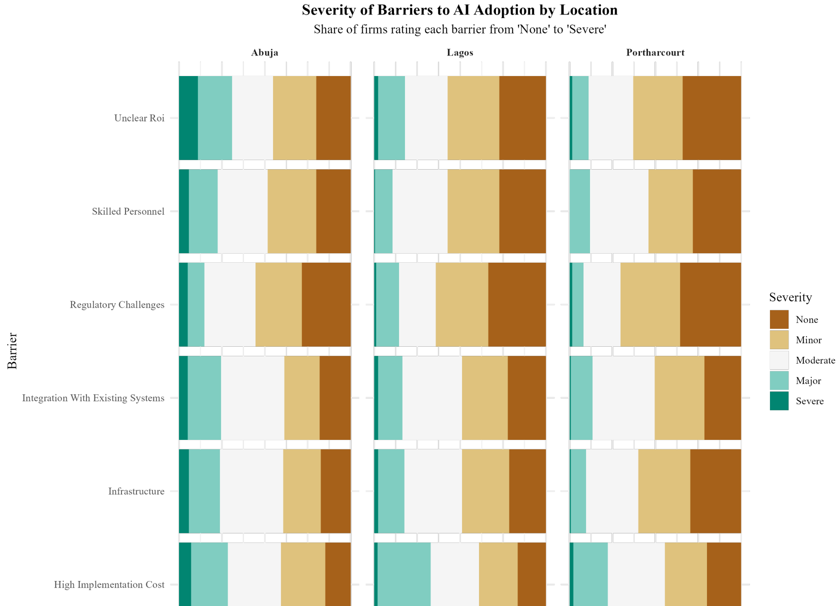
Foreign vs. Domestic Firms: A Tale of Two Models



Contrasting Patterns:

- Foreign firms: Higher absolute AI employment and higher AI intensity (as % of workforce)
- Potentially different adoption strategies:
  - Foreign: Scale advantages
  - Domestic: Focused experimentation

The Barrier Landscape



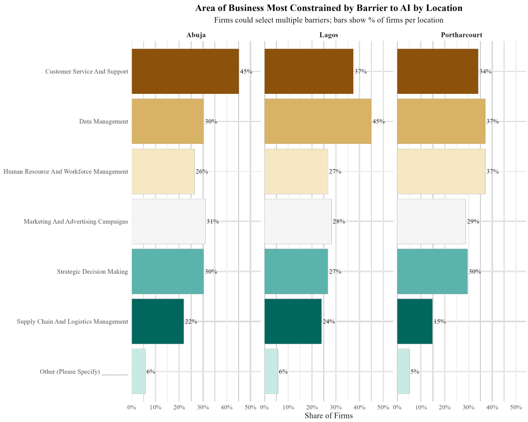
ROI Uncertainty: The Core Challenge

Survey Evidence:

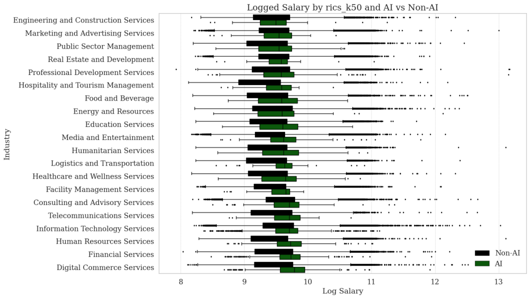
- **52%** cite "unclear ROI" as major barrier
- Lack of industry benchmarks
- No visible success stories in peer firms
- Difficulty measuring AI impact

Structural Issue:

- Not just firm-level capability gap
- Sector-wide information asymmetry
- Network effects in adoption



Evidence from Wage Premiums



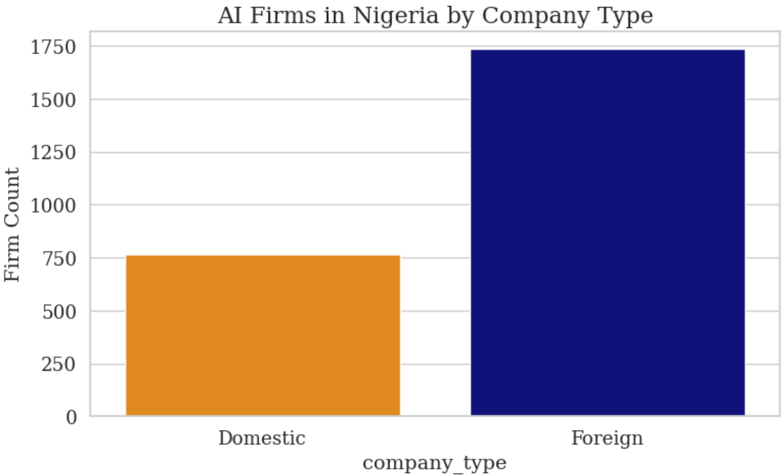
Heterogeneous Returns:

- Finance: **+20.4%** premium
- Technology: **+18.7%**
- Manufacturing: **+8.3%**
- Retail: **+3.2%**
- Public Service: **+1.1%**

Interpretation

Returns to AI vary dramatically by sector complementarities

The Productivity Puzzle



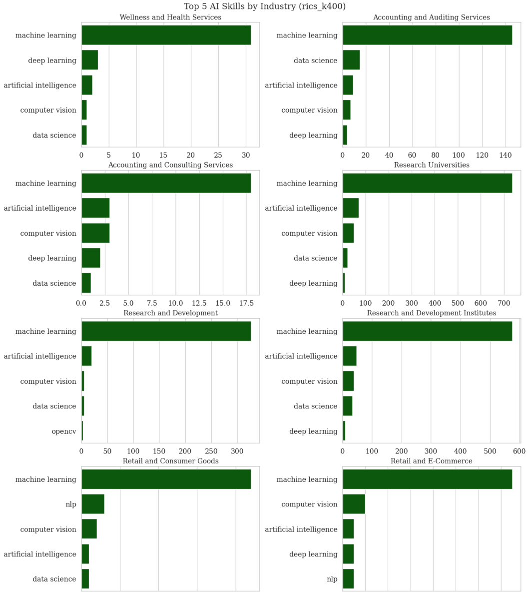
Foreign Firms:

- 10pp ↑ AI share → 6.5% wage ↑
- Clear returns to scale

Domestic Firms:

- 10pp ↑ AI share → 1.4% wage ↑ (n.s.)
- Limited productivity gains

Skills and Capabilities by Industry

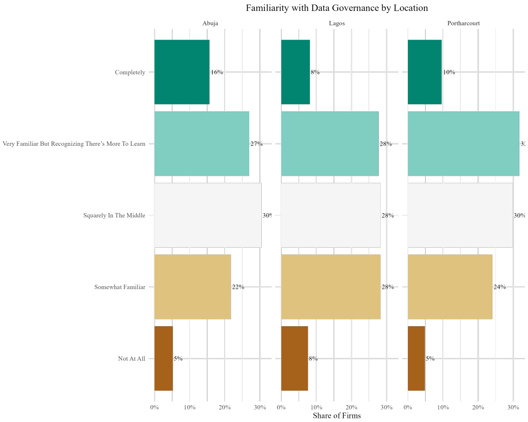


Industry-Specific Patterns:

- **Finance:** Advanced ML, risk modeling
- **Tech:** Full stack AI/ML, cloud platforms
- **Retail:** Basic analytics, chatbots
- **Manufacturing:** Computer vision, IoT

$\Rightarrow$ Successful adoption requires sector-specific capabilities

The Governance Gap



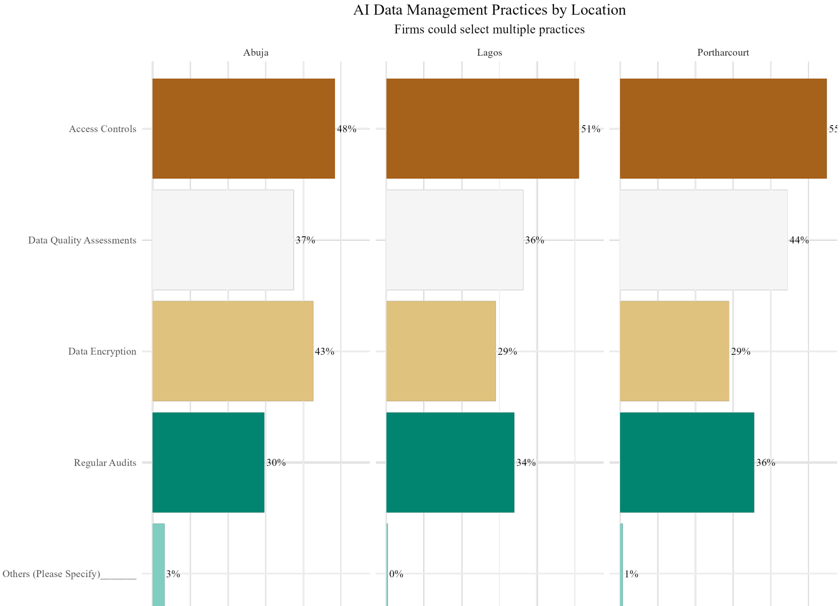
Key Findings:

- **61%** have low/no familiarity with data governance
- Only **23%** implement comprehensive data protection
- **18%** have AI ethics guidelines

Risk

Governance lags adoption, creating vulnerabilities

Data Management Practices



Key Policy Challenges

1. **Infrastructure Gaps**
   - Digital connectivity
   - Cloud computing access
   - Data center capacity
2. **Human Capital**
   - AI skills shortage
   - Limited training programs
   - Brain drain risks
3. **Market Failures**
   - Information asymmetries
   - Coordination failures
   - Missing demonstration effects
4. **Regulatory Uncertainty**
   - Data protection frameworks
   - AI governance standards
   - Cross-border data flows

Policy Recommendations

1. **Targeted Infrastructure Investment**
   - Priority sectors with high AI returns
   - Shared computing resources for SMEs
   - Public-private partnerships

2. **Skills Development Ecosystem**
   - Mid-career AI training programs
   - Industry-specific curricula
   - Retention incentives for AI talent

3. **Demonstration Projects**
   - Visible success stories in domestic firms
   - Sector-specific AI pilots
   - Knowledge sharing platforms

4. **Enabling Regulatory Framework**
   - AI services trade facilitation
   - Data governance standards
   - Innovation sandboxes

Future Research: Scaling to Continental Level

Motsepe Grant Research Program (2025-2027)

- **Geographic expansion:** Cairo, Johannesburg, Lagos, Nairobi
- **Novel measurement approach:**
  - Direct adoption (internal AI development)
  - Indirect adoption (AI services procurement)
- **Data integration:**
  - Commercial databases (Orbis, S&P Global)
  - Startup ecosystem (PitchBook, Harmonic AI)
  - Primary surveys and case studies

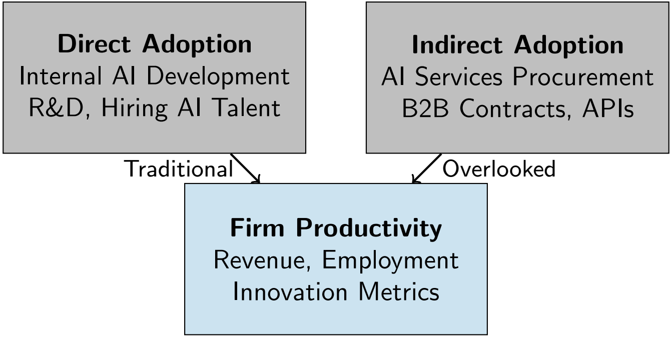
Research Timeline

Phase 1 (6 months):
Database integration

Phase 2 (10 months):
Primary data collection

Phase 3 (8 months):
Analysis & dissemination

Measuring AI Services Trade



Key Innovation: Capture AI adoption through services trade. Critical for African context where outsourcing may dominate

Key Findings

1. **Adoption Paradox:** High adoption rates (74-77%) but low intensity
   - Surface-level experimentation, not transformation
   - Concentrated in customer-facing functions
2. **Complementarity Matters:** Success requires bundled investments
   - Human capital, infrastructure, organization
   - Large firms 2.3x more likely to invest comprehensively
3. **ROI Uncertainty is Structural:** Not just a capability issue
   - Dramatic sectoral variation (1-20% wage premiums)
   - Foreign firms capture value; domestic firms struggle
4. **Governance Gap:** Adoption outpaces responsible AI practices

Implications for Theory and Practice

Theoretical Contributions:

- Extends GPT adoption to African context
- Highlights services trade channel
- Documents complementarity constraints
- Reveals sector-specific returns

Practical Implications:

- Target infrastructure to high-ROI sectors
- Create demonstration effects
- Support ecosystem development
- Enable AI services trade

Central Message: Policy must address ecosystem constraints, not just technology access

The Path Forward

From Experimentation to Transformation

Short Term

- Demonstration projects
- Skills programs
- Infrastructure gaps

Medium Term

- Sector strategies
- Services trade
- Data governance

Long Term

- Innovation ecosystem
- Regional integration
- Global competitiveness

Window of opportunity: Act now to avoid new digital divides

Thank You!

Questions & Discussion

Contact:

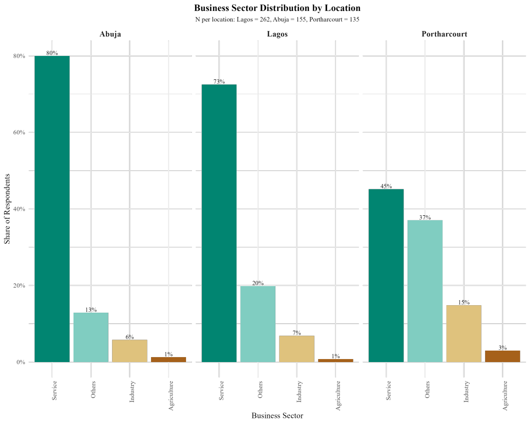
eiyoha@hbs.edu
an1490@nyu.edu
conyekwena@cseaafrica.org

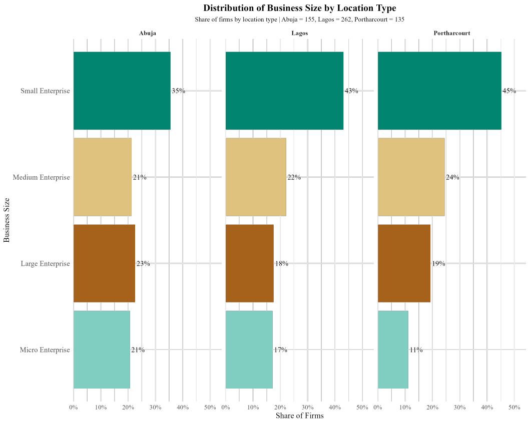
Project Website:

`www.africaai.org` (coming soon)

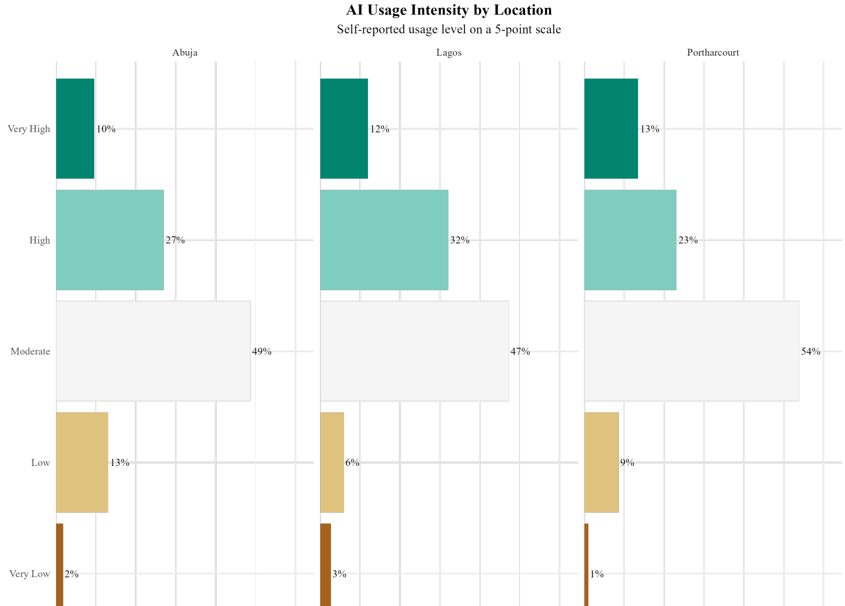
Additional Slides

Sample Demographics

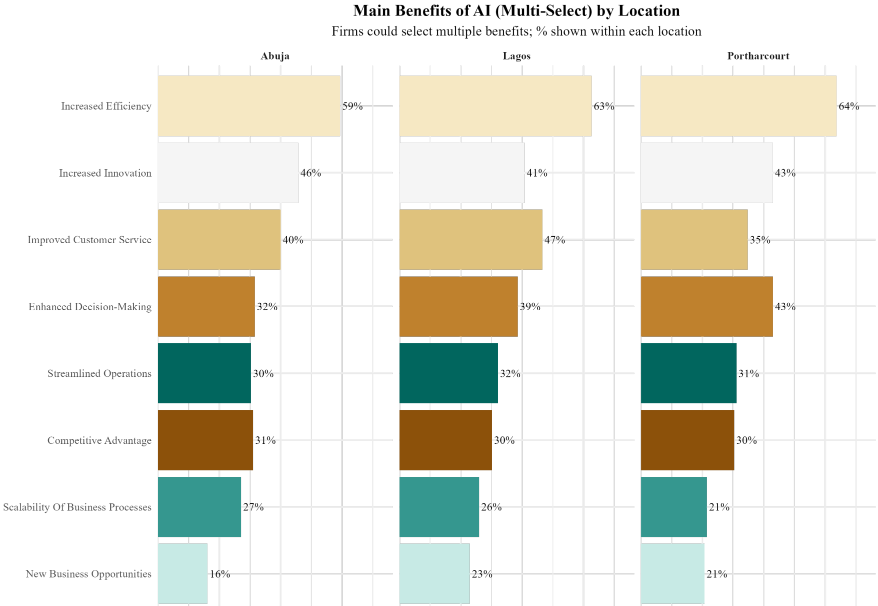




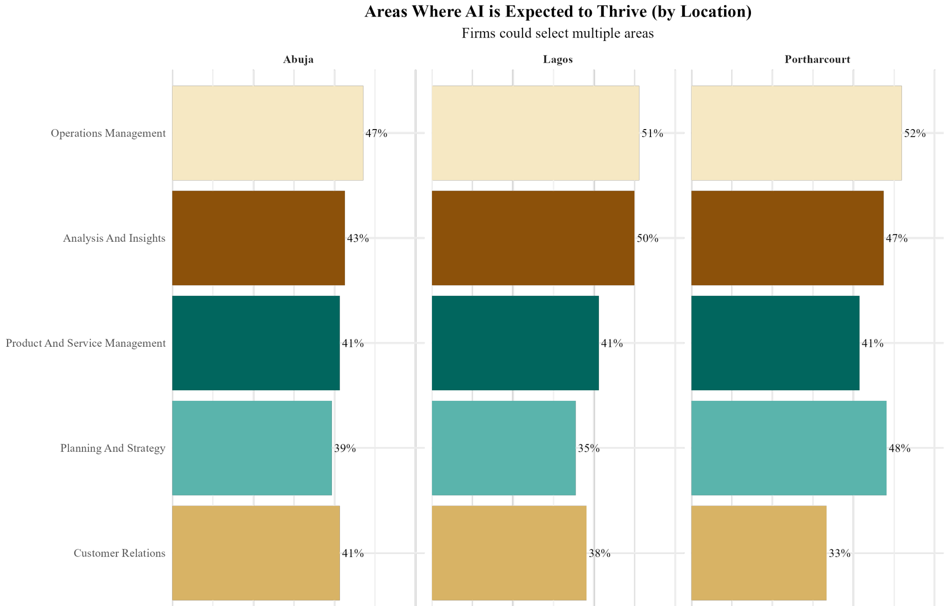
AI Usage Intensity Details



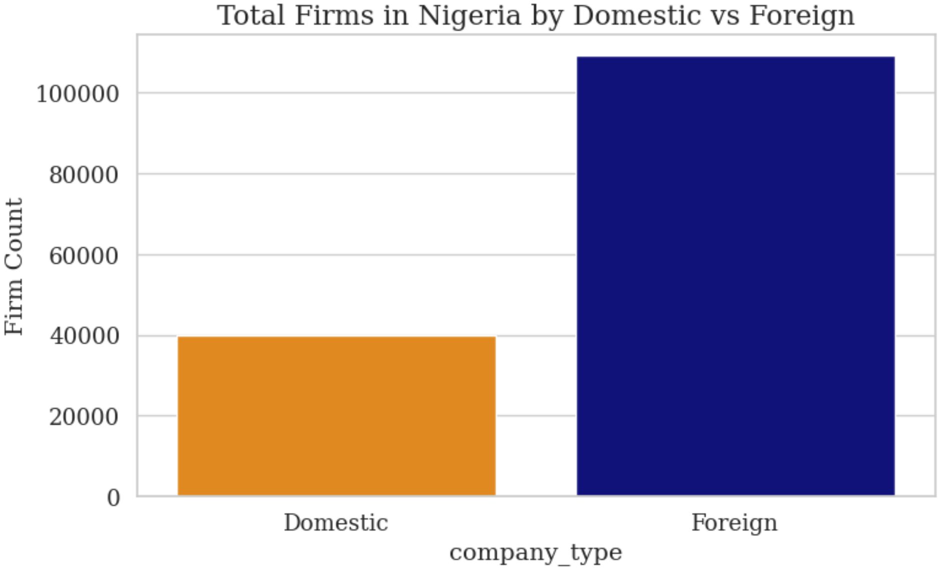
Perceived Benefits of AI



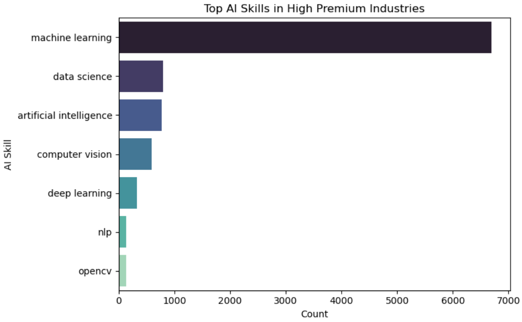
AI's Expected Impact Areas



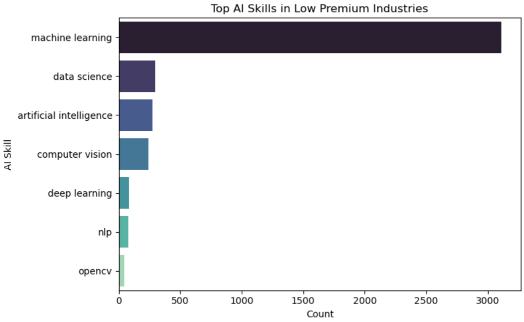
Industry Distribution of AI Workers



AI Skills in High vs Low Premium Industries

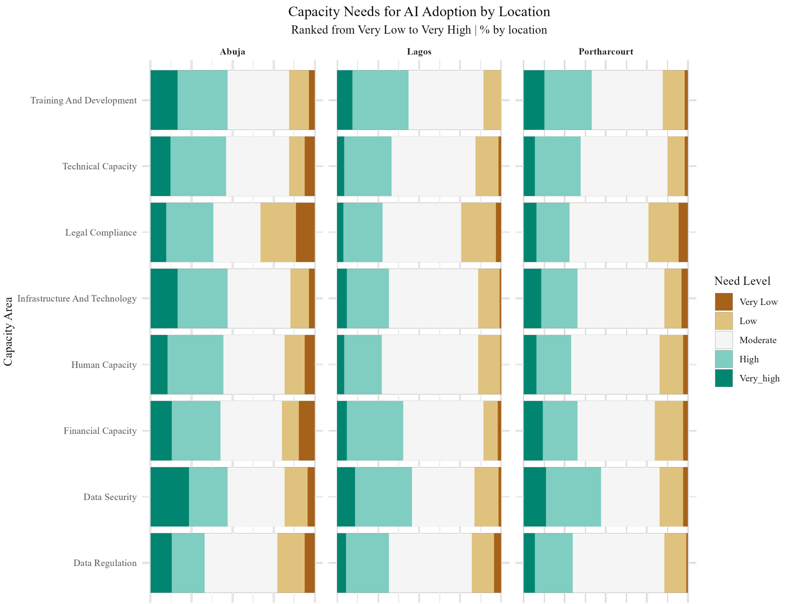


High Premium Industries

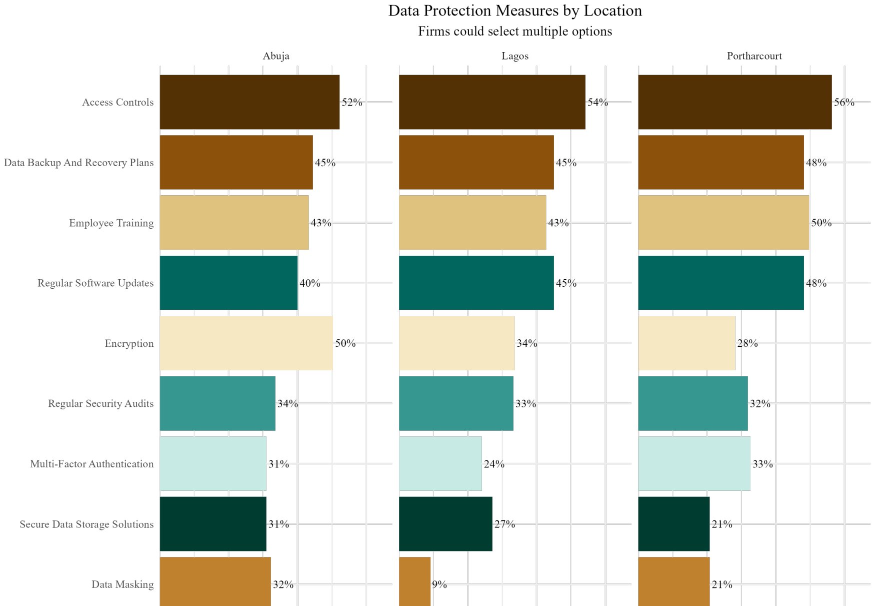


Low Premium Industries

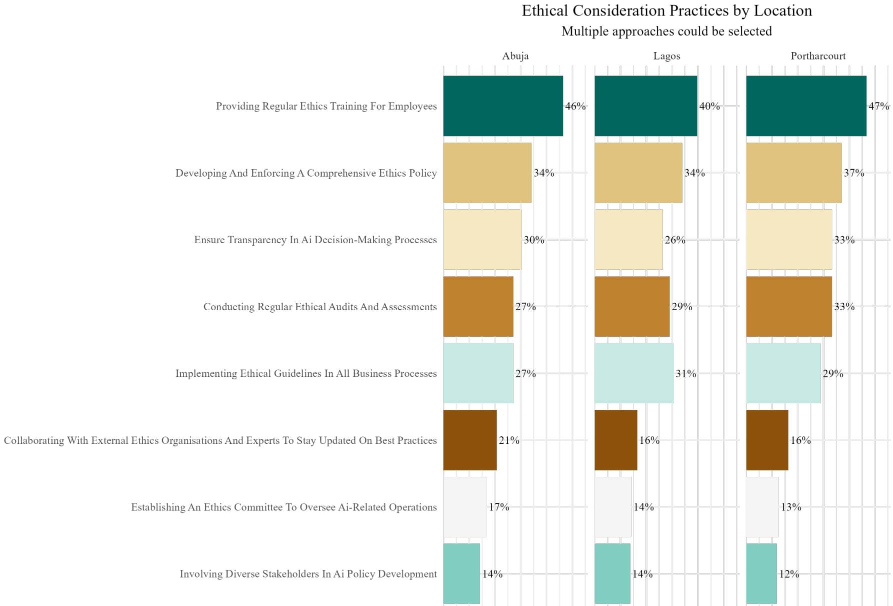
Capacity Building Needs



Data Protection Measures



Ethical AI Practices



Robustness Checks

Survey Data:

- Response rate: 73%
- No significant differences between early/late responders
- Balanced sample across firm characteristics

Revelio Data:

- Validated AI skill identification with manual coding (N=500)
- 92% accuracy in skill classification
- Wage premiums robust to outlier treatment

Limitations:

- Cross-sectional data limits causal inference
- Urban bias in sampling
- Self-reported adoption measures